Ankith Anil Kadagadakai

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Education

New Jersey City University *Jersey City, NJ | Jan 2024 to May 2025*
B.S Degree in Data Science/Business Analytics (GPA: 3.86)

Software Knowledge: Python with NumPy and Pandas, R (Dplyr, Plyr, ggplot2, viridis), Machine learning models (Scikit and Random forests), IBM SPSS for statistical analysis, Tableau/Power BI for Building dashboards and Data Modeling, MongoDB and SQL for database management, GIS and Spatial Analysis.

Other Skills: Research, Teamwork, Critical Thinking, and Problem Solving.

Thomas Edison State College *Trenton, NJ | Oct 2022 to Dec 2023*
A.S Degree in Business Administration (GPA: 3.08)

Software Knowledge: Microsoft Office Suite (MS Excel, MS Word, MS Access, MS PowerPoint, etc.), SQL for DBMS.

Other Skills: Quantitative decision making, Communication, Calculus.

Capstone Project

Single-Cell RNA-seq Analysis of GSE72056

Succesfully completed a comprehensive bioinformatics analysis project focused on identifying statistically significant differences in gene expression among cell types within the GSE72056 single-cell dataset using R.

• **Tools & Frameworks:** R (Seurat, Monocle3, CellChat, ggplot2, caret, random Forests, dplyr)

• **Key Tasks:**
- Preprocessed and normalized single-cell RNA-seq data.
- Clustered cells and validated types using known markers.
- Applied Random Forest for malignant vs. non-malignant classification.
- Conducted trajectory inference using Monocle3.
- Performed intercellular communication analysis via CellChat.
- Visualized PCA and UMAP projections for heterogeneity insights.

• **Outcomes:**
- Identified differentially expressed genes between cell types.
- Built validated ML models for cell classification.

Work Experience

Student Assistant

New Jersey City University *Jersey City, NJ | November 2024 to Present*

- Developed and implemented GIS projects, including data collection, analysis, mapping, and visualization
- Created custom maps and spatial analyses to support decision-making processes within the organization
- Achieved cost savings through effective utilization of open-source software alternatives
- Stayed up-to-date with emerging trends in GIS technology through continuous learning initiatives
- Managed large datasets using geographic information systems (GIS) software for mapping purposes

Research Presentation

Presented GIS-based urban history research at the NEDSI (Northeast Decision Sciences Institute) Conference 2025.

Project Title: Digitizing Historical Logistics Industrial Maps and Studying the Growth and Development of Jersey City Ports Post World War

- Utilized GIS tools and Blender to digitize historical maps from the 1950s and 1960s and create 3D models representing changes in port infrastructure, railways, and urban redevelopment.
- Analyzed the decline of freight and passenger railway services and industrial sectors, highlighting containerization and economic shifts.
- Integrated archival research and aerial map analysis to understand urban change and development patterns.
- Supported by NJCU's Digital Ethnic Futures Consortium (DEFCon) and STEM department; guided by Professor EunSu Lee.

Certifications and Licenses

- ArcGIS Basics – October 2024 to Present
- GIS Basics – October 2024 to Present
- Google Data Analytics – August 2024 to Present
- Power BI for Beginners – July 2024 to Present
- Introduction to SQL – April 2024 to Present